

Identification and Anatomy of Sandwich Attacks on Solana Jito

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Working Paper

1 Sandwich-Attack Identification

Our dataset is the join of Jito bundle metadata and Helius enriched transaction data, comprising 3,783,142 transactions across 1,260,686 bundles. Each transaction supplies wallet-level flow through tokenTransfers and nativeTransfers.

For every bundle of length ≥ 3 , we enumerate ordered triples (i, k, j) and construct the wallet–asset signed-flow vector $\Delta_x(w, m) = \text{inflow}_x(w, m) - \text{outflow}_x(w, m)$ per transaction. A wallet W is flagged as a candidate attacker if there exists an asset pair (b, q) such that W executes mutually reversed swaps on (b, q) between x_i and x_j (i.e., $\Delta_{x_i}(W, b)$ and $\Delta_{x_j}(W, b)$ have opposite signs and the quote flips correspondingly). We deliberately do *not* require both legs to share a feePayer: in our sample 99.7% of attackers use a dedicated tipper wallet, and any signer-based pre-filter would discard the overwhelming majority of true positives.

Each candidate (W, i, k, j) must additionally pass three economic filters: (i) **symmetry**, $|Q_f - Q_b| / \max(|Q_f|, |Q_b|) \leq 0.10$, ruling out one-sided liquidations and inventory dumps; (ii) **victim direction**: at least one wallet $V \neq W$ in some intermediate x_k has a quote-asset flow aligned with W 's front leg; (iii) **positive base profit**. Two cleaning rules remove residual artefacts: R1 drops bundles whose flagged attacker is also a victim signer or aligned wallet (self-attacks); R2 blacklists wallets that appear as candidate attacker in ≥ 200 bundles with balanced direction ($\min(b, s) / \max(b, s) \geq 0.30$) and mean bundle profit within ± 0.01 SOL—the empirical signature of two-sided market makers rather than opportunistic attackers.

The pipeline identifies 11,591 **confirmed Pattern-A sandwich attacks**, spanning 12,916 unique slots and 161 unique attacker wallets; 5,060 (38.4%) realise positive net SOL profit. This count is approximately *two orders of magnitude* larger than previously reported Pattern-A volumes on Solana, the gap being attributable almost

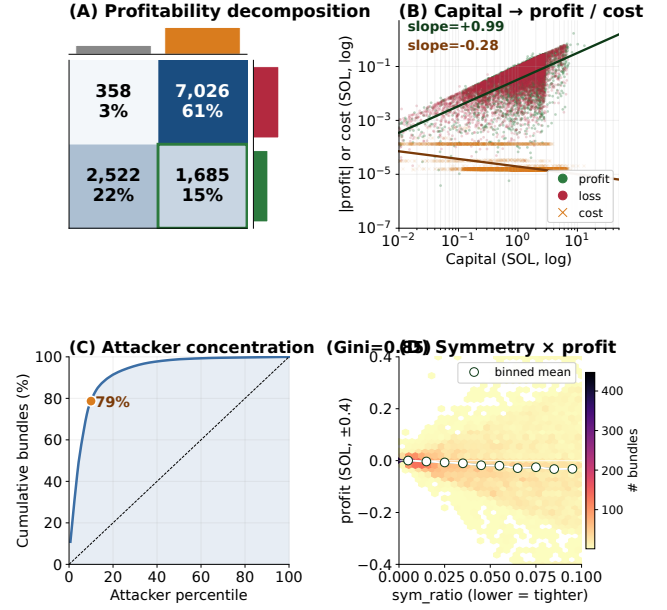


Figure 1: Four-axis anatomy of 11,591 confirmed sandwiches.

entirely to dropping the implicit feePayer-equality assumption.

Figure 1 dissects the confirmed sample along four dimensions. Panel (A) is the joint decomposition of price-edge versus net-SOL profitability: although 75% of attackers correctly position on the price spread, only 36% realise positive net SOL profit; the doubly-profitable cell accounts for just 14.5% of attempts. Most “price-edge wins” are erased by pool slippage and residual base-token inventory. Panel (B) projects both $|profit|$ and $tip + fee$ onto the capital axis on log–log scales: the profit slope is ≈ 0.99 (nearly linear in capital), whereas the cost slope is ≈ -0.28 (essentially flat). Marginal cost is close to zero, granting capital-rich attackers a structural scale advantage. Panel (C) shows that the market is highly concentrated (Gini = 0.78, top 1% executes 11% and top 10% executes 79%), while two insets reveal that 86% of attacks are front-buyer attacks and

99.2% of bundles target only one victim. Panel (D) plots the joint density of execution symmetry (sym_ratio) and SOL profit: $\text{sym_ratio} < 0.02$ yields positive expected profit, whereas $\text{sym_ratio} > 0.05$ is loss-making

on average—in the absence of bidding wars, what separates winners from losers is execution precision rather than tip aggression.